Jared F. Miller

Optimization · Control · Power Electronics

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Education

Northeastern University

Boston, MA, USA

Ph.D. in Electrical and Computer Engineering *Sept. 2018 - May 2023*

- Communications, Control, and Signal Processing (CCSP)
- Advised by Prof. Mario Sznaier
- Thesis Title: "Safety Quantification for Nonlinear and Time-Delay Systems using Occupation Measures" [link] [interview]
- Thesis Committee: Octavia Camps, Didier Henrion (LAAS-CNRS), Bahram Shafai, Eduardo Sontag, Mario Sznaier
- GPA: 4.0 (4.0 Scale)

M.S. in Electrical and Computer Engineering *Sept. 2015 - May 2018*

- Communications, Control, and Signal Processing (CCSP)
- GPA: 3.852 (4.0 Scale)

B.S. in Electrical Engineering *Sept. 2013 - May 2018*

- Minor in Mathematics
- GPA: 3.865 (4.0 Scale), Summa cum Laude

Research Publications

Journal Papers (published)

1. Jared Miller, Fabian Jakob, Carsten Scherer, and Andrea Iannelli. Analysis and synthesis of switched optimization algorithms. *Nonlinear Analysis: Hybrid Systems*, 62:101788, 2026. [link]
2. Jared Miller and Mario Sznaier. Quantifying the safety of trajectories by the data corruption needed to crash. *SIAM Journal on Control and Optimization*, 64(4):2204–2223, 2026. [link]
3. Jared Miller, Matteo Tacchi, Didier Henrion, and Mario Sznaier. Unsafe probabilities and risk contours for stochastic processes using convex optimization. *Automatica*, 188:112939, 2026. [link]
4. Jared Miller, Jie Wang, and Feng Guo. Sparse polynomial matrix optimization. *SIAM Journal on Optimization*, 36(2):503–533, 2026. [link]
5. Jared Miller, Tianyu Dai, Mario Sznaier, and Bahram Shafai. Data-Driven Control of Switched Linear Positive Systems using Robust Convex Optimization. *IEEE Transactions on Automatic Control*, pages 1–8, 2026. [link], Early Access
6. Jared Miller and Mario Sznaier. Peak estimation of hybrid systems with convex optimization. *Automatica*, 183:112641, 2026. [link]
7. Tooba Imtiaz, Morgan R Kohler, Jared F Miller, Zifeng Wang, Masih Eskandar, Mario Sznaier, Octavia Camps, and Jennifer Dy. SAIF: Sparse adversarial and imperceptible attack framework. *Transactions on Machine Learning Research*, 2025. [link]
8. Jared Miller and Mario Sznaier. Analysis of Input-Affine Dynamical Systems using Parameterized Robust Counterparts. *IEEE Transactions on Automatic Control*, 70(8):5413–5428, 2025. [link]
9. Jared Miller, Matteo Tacchi, Mario Sznaier, and Ashkan Jasour. Convex Computation of Value-at-Risk Bounds for Stochastic Processes. *IEEE Transactions on Automatic Control*, 70(6):3936–3951, 2025. [link]
10. Jared Miller, Tianyu Dai, and Mario Sznaier. Robust Data-Driven Control of Discrete-Time Linear Systems with Errors in Variables. *IEEE Transactions on Automatic Control*, 70(2):947–962, 2025. [link]
11. Jared Miller and Roy S. Smith. Peak estimation of rational systems using convex optimization. *European Journal of Control*, page 101088, 2024. [link] (2024 European Control Conference Special Issue)
12. Jared Miller, Tianyu Dai, and Mario Sznaier. Data-Driven Superstabilizing Control Under Quadratically-Bounded Errors-in-Variables Noise. *IEEE Control Systems Letters*, 8:1655–1660, 2024. [link]
13. Jian Zheng, Jared Miller, and Mario Sznaier. Data-Driven Safe Control of Discrete-Time Non-Linear Systems. *IEEE Control Systems Letters*, 8:1553–1558, 2024. [link]
14. Jared Miller and Mario Sznaier. Bounding the Distance to Unsafe Sets with Convex Optimization. *IEEE Transactions on Automatic Control*, 68(12):7575 – 7590, 2023. [link]
15. Jian Zheng, Tianyu Dai, Jared Miller, and Mario Sznaier. Robust Data-Driven Safe Control using Density Functions. *IEEE Control Systems Letters*, 7:2611–2616, 2023. [link] (LCSS-CDC)
16. Jared Miller and Mario Sznaier. Data-Driven Gain Scheduling Control of Linear Parameter-Varying Systems using Quadratic Matrix Inequalities. *IEEE Control Systems Letters*, 7:835–840, 2022. [link] (LCSS-ACC)
17. Jared Miller, Yang Zheng, Mario Sznaier, and Antonis Papachristodoulou. Decomposed structured subsets for semidefinite and sum-of-squares optimization. *Automatica*, 137:110–125, 2022. [link]
18. J. Miller, D. Henrion, and M. Sznaier. Peak Estimation Recovery and Safety Analysis. *IEEE Control Systems Letters*, 5(6):1982–1987, 2021. [link] (LCSS-ACC)
19. Jared Miller, Muhammad Ali Al-Radhawi, and Eduardo Daniel Sontag. Mediating Ribosomal Competition by Splitting Pools. *IEEE Control Systems Letters*, 5(5):1555–1560, 2021. [link] (LCSS-ACC)

Journal Papers (accepted but not yet published)

1. Jared Miller, Mattia Bianchi, and Florian Dörfler. Convergence and Robustness Bounds for Distributed Asynchronous Shortest-Path, 2026. [link], (Transactions on Control of Network Systems)
2. Jared Miller. Incremental Peak-to-Peak Gain Computation and Regulation of Discrete-Time Positive Luré Systems using Linear Programming , 2026. [link], (Automatica, Technical Note)

Journal Papers (submitted)

1. Jared Miller, Carsten Scherer, Fabian Jakob, and Andrea Iannelli. Structure, Analysis, and Synthesis of First-Order Algorithms, 2026. [link]
2. Niklas Schmid, Jared Miller, Tristan Zeller, Marta Fochesato, Tobias Sutter, and John Lygeros. Joint Chance Constrained Safe-Optimal Control, 2026. [link]
3. Chiara Meroni, Jared Miller, and Mauricio Velasco. Approximation of starshaped sets using polynomials, 2025. [link]
4. Jared Miller, Chiara Meroni, Matteo Tacchi, and Mauricio Velasco. Maximizing Slice-Volumes of Semialgebraic Sets using Sum-of-Squares Programming, 2024. [link]
5. Jared Miller, Niklas Schmid, Matteo Tacchi, Didier Henrion, and Roy S. Smith. Peak Time-Windowed Risk Estimation of Stochastic Processes, 2024. [link]
6. Didier Henrion, Jared Miller, and Mohab Safey El Din. Algebraic Proofs of Path Disconnectedness using Time-Dependent Barrier Functions, 2024. [link]

Conference Proceedings (published)

1. Jared Miller and Petros Karamanakos. Optimal pulse patterns through a hybrid optimal control perspective. In *2025 IEEE 64th Conference on Decision and Control (CDC)*, pages 6565–6572, 2025. [link]
2. Jared Miller, Jaap Eising, Florian Dörfler, and Roy S. Smith. Data-driven structured robust control of linear systems. In *2025 European Control Conference (ECC)*, pages 96–101, 2025. [link]
3. Pierluigi Francesco De Paola, Jared Miller, Alessandro Borri, Alessia Paglialonga, and Fabrizio Dabbene. Robust control-driven counterfactual generation for uncertain systems. *11th IFAC Symposium on Robust Control Design ROCOND 2025*, 59(16):116–121, 2025. [link]
4. Jared Miller, Jian Zheng, Mario Sznaier, and Chris Hixenbaugh. Data-driven superstabilization of linear systems under quantization. In *2024 American Control Conference (ACC)*, pages 1146–1151, 2024. [link]
5. Mohamed Abdalmoaty, Jared Miller, Mingzhou Yin, and Roy S. Smith. Frequency-domain identification of discrete-time systems using sum-of-rational optimization. In *20th IFAC Symposium on System Identification SYSID 2024*, volume 58, pages 121–126, 2024. [link]
6. Jared Miller, Niklas Schmid, Matteo Tacchi, Didier Henrion, and Roy S. Smith. Peak time-windowed mean estimation using convex optimization. In *2024 IEEE 63rd Conference on Decision and Control (CDC)*, pages 703–708, 2024. [link]
7. Arya Honarpisheh, Rajiv Singh, Jared Miller, and Mario Sznaier. Identification of low order systems in a loewner framework. In *20th IFAC Symposium on System Identification SYSID 2024*, volume 58, pages 199–204, 2024. [link]
8. Jared Miller, Matteo Tacchi, Mario Sznaier, and Ashkan Jasour. Peak Value-at-Risk Estimation for Stochastic Differential Equations Using Occupation Measures. In *2023 62nd IEEE Conference on Decision and Control (CDC)*, pages 4836–4842, 2023. [link]
9. Jared Miller, Tianyu Dai, Mario Sznaier, and Bahram Shafai. Data-Driven Control of Positive Linear Systems using Linear Programming. In *2023 62nd IEEE Conference on Decision and Control (CDC)*, pages 1588–1594, 2023. [link]
10. Jared Miller, Milan Korda, Victor Magron, and Mario Sznaier. Peak Estimation of Time Delay Systems Using Occupation Measures. In *2023 62nd IEEE Conference on Decision and Control (CDC)*, pages 5294–5300, 2023. [link]
11. Jared Miller, Tianyu Dai, and Mario Sznaier. Superstabilizing Control of Discrete-Time ARX Models under Error in Variables. In *22nd IFAC World Congress*, volume 56, pages 2444–2449, 2023. [link] (Finalist for IFAC Young Author Award)
12. Jared Miller and Mario Sznaier. Bounding the Distance of Closest Approach to Unsafe Sets with Occupation Measures. In *2022 61st IEEE Conference on Decision and Control (CDC)*, pages 5008–5013, 2022. [link]
13. Jared Miller, Tianyu Dai, and Mario Sznaier. Data-Driven Superstabilizing Control of Error-in-Variables Discrete-Time Linear Systems. In *2022 61st IEEE Conference on Decision and Control (CDC)*, pages 4924–4929, 2022. [link] (Outstanding Student Paper Award)
14. Filip Bečanović, Jared Miller, Vincent Bonnet, Kosta Jovanović, and Samer Mohammed. Assessing the Quality of a Set of Basis Functions for Inverse Optimal Control via Projection onto Global Minimizers. In *2022 IEEE 61st Conference on Decision and Control (CDC)*, pages 7598–7605, 2022. [link]
15. Jared Miller and Mario Sznaier. Facial Input Decompositions for Robust Peak Estimation under Polyhedral Uncertainty. *IFAC-PapersOnLine*, 55(25):55–60, 2022. [link] (IFAC Young Author Award)
16. Jared Miller, Didier Henrion, Mario Sznaier, and Milan Korda. Peak Estimation for Uncertain and Switched Systems. In *2021 60th IEEE Conference on Decision and Control (CDC)*, pages 3222–3228, 2021. [link] (Outstanding Student Paper Award)
17. J. Miller, R. Singh, and M. Sznaier. MIMO System Identification by Randomized Active-Set Methods. In *2020 59th IEEE Conference on Decision and Control (CDC)*, pages 2246–2251, 2020. [link]
18. Jared Miller, Yang Zheng, Mario Sznaier, and Antonis Papachristodoulou. Decomposed Structured Subsets for Semidefinite Optimization. In *2020 21st IFAC World Congress*, 2020. [link]
19. Chieh Wu, Jared Miller, Yale Chang, Mario Sznaier, and Jennifer Dy. Solving Interpretable Kernel Dimensionality Reduction. In H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 32, pages 7915–7925. Curran Associates, Inc., 2019. [link]
20. J. Miller, Y. Zheng, B. Roig-Solvas, M. Sznaier, and A. Papachristodoulou. Chordal Decomposition in Rank Minimized Semidefinite Programs with Applications to Subspace Clustering. In *2019 IEEE 58th Conference on Decision and Control (CDC)*, pages 4916–4921, 2019. [link]
21. B. Taskazan, J. Miller, U. Inyang-Udoh, O. Camps, and M. Sznaier. Domain Adaptation Based Fault Detection in Label Imbalanced Cyberphysical Systems. In *2019 IEEE Conference on Control Technology and Applications (CCTA)*, pages 142–147, 2019. [link]
22. J. Miller and B. Shafai. A Model of Heave Dynamics for Bagged Air Cushioned Vehicles. In *2019 IEEE Conference on Control Technology and Applications (CCTA)*, pages 976–981, 2019. [link]

Conference Proceedings (accepted)

1. Jared Miller, Nicolás Estay, Petros Karamanakos, and Alejandro Angulo. Symmetry-Reduced Algebraic Selective Harmonic Elimination for Cascaded H-Bridge Converters with Unequal DC Source Levels, 2026

Preprints

1. Jared Miller, Petros Karamanakos, and Tobias Geyer. Bounding the Minimal Current Harmonic Distortion in Optimal Modulation of Single-Phase Power Converters, 2025. [link]

Seminars (Selected)

1. "Structure, Analysis, and Synthesis of First-Order Algorithms," March 27, 2026, KTH, Stockholm, SE [link].
2. "Are optimal pulse patterns optimal? Bounding minimal harmonic distortion of DC/AC power converters," November 28, 2025, Control Systems Talks, FAU Erlangen-Nürnberg, DE [link].
3. "Analysis and Control of Input-Affine Systems using Parameterized Robust Counterparts," August 19-23, 2024, 26th International Symposium on Mathematical Theory of Networks and Systems, Cambridge, UK. [link]
4. "Frequency-Domain Identification of Discrete-Time Systems using Sum-of-Rational Optimization," August 19-23, 2024, 26th International Symposium on Mathematical Theory of Networks and Systems, Cambridge, UK. [link]
5. "Going beyond ODE systems with the moment-SOS hierarchy," July 29, 2024, Polynomial Optimization for Nonlinear Dynamics: Theory, Algorithms, and Applications, Oberwolfach, DE. [link]
6. "Risk Analysis of Stochastic Processes using Occupation Measures," April 11, 2024, H-CODE Seminar Series, Laboratory of Signals and Systems (L2S), Paris-Saclay, FR. [link]
6. "Maximizing the Slice-Volume of Semialgebraic Sets using Sum-of-Squares Programming," March 11-14, 2024, Moments and Polynomials: Applications and Theory (MoPAT24), Konstanz, DE. [link]
7. "Data-Driven Safety Quantification using Robust Optimization," February 16, 2024, Institute for Systems Theory and Automatic Control (IST), University of Stuttgart, Stuttgart, DE. [link].
8. "Analysis and Control of Time-Delay Systems Using Polynomial Optimization", May 14, 2023, MS14 Studying Dynamics using Polynomial Optimization Tools, SIAM Conference on Dynamical Systems. [link]
9. "Data-Driven Control under Input and Measurement Noise", April 9, 2023, Oden Institute Seminar, UT Austin. [link]
10. "Safety Analysis for Nonlinear and Time-Delay Systems using Occupation Measures", April 3, 2023, PhD Thesis Defense, Northeastern University. [link].
11. Tutorials about Interior Point Methods, Polynomial Optimization, Frank-Wolfe algorithms and variations, and SDP approximations, May 16-20, Sparsity and Big Data in Control, Systems Identification, and Machine Learning, European Embedded Control Institute.
12. "Bounding distances to unsafe sets", April 14, 2022, Conic Linear Optimization for Computer-Assisted Proofs, Mathematisches Forschungsinstitut Oberwolfach (MFO). [link]
13. "Data-Driven Peak and Reachability Set Estimation", May 25, 2021, MS112 Methods of Learning Dynamical Systems for Control, SIAM Conference on Dynamical Systems. [link]
14. "Exploiting Structure in Rank-Constrained and Approximated Semidefinite Programs", December 19, 2019, TISEM Operations Research Seminar, Tilburg University. [link]

Poster Sessions (without Conference Proceedings)

1. "Are Optimal Pulse Patterns Optimal? Bounding harmonic distortion in power converters." September 2, 2025. Swiss CLOCK summit, Engelberg (Best Poster Award, juried). [link]
2. "Quantifying Trajectory Safety by the Minimum Data Corruption Needed to Crash." September 30-October 2, 2024. Symposium on Systems Theory in Data and Optimization. Universität Stuttgart. [link]
3. "Risk Analysis of Stochastic Processes using Polynomial Optimization." November 13, 2023, Future Trends in Polynomial Optimization, LAAS-CNRS, Toulouse, FR. [link]
4. "Frequency Domain Identification via Sum-of-Rational Optimization." September 25, 2023, European Research Network System Identification (ERNSI), Stockholm, SE. [link]
5. "Safety Analysis and Control using Measures." April 13, 2023, RISE 2023, Northeastern University. [link]
6. "Safety Analysis and Control using Measures." February 27, 2023, PhD Research Expo, Northeastern University. [link]
7. "Diameter Constrained Minimum Spanning Graphs." January 31, 2023, Current Themes of Discrete Optimization: Boot-camp for early-career researchers, ICERM. [link]
8. "Safety Analysis using Distance Estimation and Measures." August 24, 2022. CLEVR-AI MURI Yearly Review Meeting, Northeastern University. [link]
9. "Exploiting SDP Structure Yields Tighter Approximations." April 9, 2020. RISE, Northeastern University (remote). [link]
10. "Exploiting SDP Structure Yields Tighter Approximations." February 24, 2020. IPAM Control, Learning and Optimization workshop, University of California, Los Angeles. [link]
11. "Chordal Decompositions in Rank Minimized SDPs." May 30-31, 2019. Learning for Decision and Control (L4DC), Massachusetts Institute of Technology. [link]
12. "Chordal Decompositions in Rank Minimized SDPs." May 10, 2019. New England Machine Learning Day, Northeastern University. [link]
13. "Scattered data interpolation through B-spline wavelets and the Elastic Net." April 14, 2017. RISE, Northeastern University. [link]
14. "A parallelized Python-based Multi-Point Thomson Scattering analysis in NSTX-U." October 29, 2014. 56th Annual APS Plasma Physics Conference, New Orleans. [link]

Experience

University of Stuttgart — *Stuttgart, DE*
Postdoctoral Researcher, Mathematical Systems Theory, group of Prof. Carsten Scherer — *February 2025 - Present*

ETH Zürich — *Zürich, CH*
Postdoctoral Researcher, Automatic Control Lab (IfA), Forschungsgruppe Prof. Roy Smith — *August 2023 - December 2024*

Northeastern University — *Boston, MA, USA*
Postdoctoral Researcher, Robust Systems Lab, Controls Group — *May 2023 - July 2023*

Northeastern University — *Boston, MA, USA*
Research Assistant, Robust Systems Lab, Controls Group — *Jul. 2017 - April 2023*

Laboratory for Analysis and Architecture of Systems (LAAS-CNRS) — *Toulouse, FR*
Chateaubriand Fellow, Decision and Optimization, Methods and Algorithms in Control (MAC) Team — *Jan. 2022 - Jul. 2022*

Paradigm Hyperloop — *Boston, MA, USA*
Control Theorist, Levitation Group — *Jul. 2017 - Dec. 2017*

ASML Holding — *Veldhoven, NL*
Co-op, Metrology Group — *Mar. 2016 - Aug. 2016*

Advanced Micro Devices (AMD) — *Boxborough, MA, USA*
Co-op, Shader Compiler Group — *Jan. 2015 - Jun. 2015*

Princeton Plasma Physics Laboratory (PPPL) — *Plainsboro Township, NJ, USA*
Intern/Part-time Contractor, Data Visualization Group — *Sep. 2012 - Feb. 2016*

Cornell High Energy Synchrotron Source (CHESS) — *Ithaca, NY, USA*
Summer Intern/Research Assistant — *Jul. 2012 - Aug. 2012*

Teaching

University of Stuttgart — *Stuttgart, DE*

Teaching Assistant — *Spring/Summer 2026*
- Robust Control

Teaching Assistant — *Fall/Winter 2025*
- Linear Matrix Inequalities

Teaching Assistant — *Fall/Winter 2025*
- Linear Control Theory

Teaching Assistant — *Spring/Summer 2025*
- Higher Mathematics 2

ETH Zürich — *Zürich, CH*

Teaching Assistant — *Fall 2024*
- System Identification (227-0689)

Teaching Assistant — *Spring 2024*
- Robust Control & Convex Optimization (227-0690-13L)

Teaching Assistant — *Fall 2023*
- System Identification (227-0689)

Northeastern University *Boston, MA, USA*

Teaching Assistant *Fall 2022*

- EECE 5644: Machine Learning and Pattern Recognition

Teaching Assistant *Fall 2021*

- EECE 7345: Big Data, Sparsity, and Control

European Embedded Control Institute *Toulouse, FR*

Teaching Assistant *May 16-20, 2022*

- Sparsity and Big Data in Control, Systems Identification and Machine Learning

Student Supervision

University of Stuttgart *Stuttgart, DE*

Masters' Thesis *Jan. 2026-Aug 2026*

- Optimization Algorithm Analysis and Design with Integral Quadratic Constraints (Lennart Dahlhues)

University of Stuttgart *Stuttgart, DE*

Masters' Thesis *Apr.-Nov. 2025*

- Planning Power Grid Upgrades to Minimize Peak-to-Peak Responses to Transient Faults (Emna Ayadi)

ETH Zürich *Zürich, CH*

Masters' Thesis *Mar.-Oct. 2024*

- Plug and Play Distributed Control of Energy Hub Networks (Cara Koepele, co-supervised with Varsha Behrunani)

Masters' Semester Project *Sept.-Dec. 2024*

- Model Predictive Control for Jointly Managing Buildings and Electric Bus Fleets under Bidirectional Charging (Aneska Heidemüller, co-supervised with Federica Bellizio and Marta Fochesato)

Masters' Semester Project *Mar.-Jun. 2024*

- Symmetry-Aware Recurrent Equilibrium Neural Networks (Shubham Gupta, co-supervised with Jiaqi Yan)

Honors & Awards

Sept. 2025 **Best Poster (juried),** Swiss CLOCK Summit *Engelberg, CH*

Jan. 2025 **Outstanding Service as Reviewer (2024 year),** IEEE L-CSS *N/A*

Jul. 2023 **Finalist for Young Author Award,** 2023 IFAC World Congress *Yokohama, JP*

Jun. 2023 **Travel Award,** 2023 American Control Conference (ACC) *San Diego, CA, USA*

Apr. 2023 **ECE Excellence in Research Award,** Northeastern University College of Engineering *Boston, MA, USA*

Jan. 2023 **Travel Award,** ICERM workshop: Current Themes of Discrete Optimization *Providence, RI, USA*

Dec. 2022 **Outstanding Student Paper Award,** 2022 61st IEEE Conference on Decision and Control *Cancún, MX*

Dec. 2022 **Travel Award,** 2022 61st IEEE Conference on Decision and Control *Cancún, MX*

Oct. 2022 **ASME DSCD Rising Star Award,** MECC 2022 (IFAC) *Jersey City, NJ, USA*

Sep. 2022 **IFAC Young Author Award,** ROCOND 2022 (IFAC) *Kyoto, JP (remote)*

Apr. 2022 **Travel Award,** MFO: Conic Linear Optimization for Computer-Assisted Proofs *Oberwolfach, DE*

Jan. 2022 **Travel Award (for Toulouse),** AFOSR FY22 International Student Exchange Program (ISEP) *Arvada, CO, USA*

Dec. 2021 **Outstanding Student Paper Award,** 2021 60th Conference on Decision and Control *Austin, TX, USA*

Apr. 2020 **Chateaubriand Fellowship,** Office of Science and Technology, Embassy of France in the USA *Toulouse, FR*

Feb. 2020 **Hosting,** IPAM: Intersections between Control, Learning and Optimization (UCLA Workshop) *LA, CA, USA*

Dec. 2019 **Travel Award for Seminar,** TISEM Seminar at Tilburg University *Tilburg, NL*

Aug. 2019 **Travel Award,** 2019 IEEE Conference on Control Technology and Applications (CCTA) *Hong Kong*

2013-2018 **Honors Program,** Northeastern University *Boston, MA, USA*

2015-2018 **Dean's List,** Northeastern University *Boston, MA, USA*

Skills

Programming Matlab (incl. Simulink), Python, Mathematica, LaTeX, Julia, C/C++

MS Office Word, Excel, PowerPoint, Publisher

Professional Organizations

Institute of Electrical and Electronics Engineers (IEEE)

Member — *Sept. 2013 - Present*

IEEE Eta Kappa Nu (HKN)

Member, Tutor — *Sept. 2014 - Present*

Society for Industrial and Applied Mathematics

Member — *Oct. 2019 - Present*

IEEE CSS Technical Committee on Robust and Complex Systems (TC-RoCS)

Member — *Sept. 2022 - Present*

IEEE CSS Technical Committee on Hybrid Systems

Member — *Jun. 2023 - Present*

IEEE CSS Technical Committee on Stochastic Systems and Controls

Member — *July 2024 - Present*

IFAC Technical Committee 2.5 Robust Control

Member — *Sept. 2022 - Present*

INFORMS

Member — *June 2023 - Present*

Professional Service

Reviewer

Automatica, IEEE Transactions on Automatic Control (TAC), Transactions on Control Systems Technology (TCST), IEEE Control Systems Letters (L-CSS), European Journal of Control (EJC), Learning for Dynamics & Control Conference (L4DC), IFAC Symposium on Robust Control Design (ROCOND), Symposium on Systems Theory in Data and Optimization (SysDO), IEEE Conference on Decision and Control (CDC), Mathematics of Operations Research, European Control Conference (ECC), Nonlinear Analysis: Hybrid Systems (NA:HS), Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences; Kybernetica, Association for the Advancement of Artificial Intelligence (AAAI)

Guest Editor

Mathematics of Control, Signals, and Systems: “Polynomial Optimization for Control, Signals and Systems”

Committee

Best Student Paper Award committee, IEEE Conference on Decision and Control, 2026.